

Ten-year response of northern hardwood stands to commercial selection cutting in southern Quebec, Canada

Eric Forget^{*}, Philippe Nolet, Frédérik Doyon, Sylvain Delagrange, Yves Jardon

Institut Québécois d'Aménagement de la Forêt Feuillue (IQAFF), 58 rue Principale, Ripon, Quebec J0V 1V0, Canada

Received 18 July 2006; received in revised form 9 February 2007; accepted 13 February 2007

Abstract

Trees sampled from 12 uneven-aged sugar maple (*Acer saccharum*, Marsh.) dominated stands in southern Quebec were measured 10 years after commercial single-tree selection cutting. Dendrochronological analysis revealed that growth response to treatment was immediate for all three species. Species-specific response was, however, stronger for American beech (*Fagus grandifolia*, Ehrh.) than for sugar maple and yellow birch (*Betula alleghaniensis*, Britton). When averaged for 5-year periods, species growth during the second 5-year period after treatment remained significantly higher than their growth prior to treatment. Computations of whole stand minimum and maximum mortality (0.04 and 0.20 m² ha⁻¹ year⁻¹) and the associated net growth (0.21 and 0.37 m² ha⁻¹ year⁻¹) were in the range of what was observed in the Quebec Government experimental blocks. Finally, at lower basal areas (i.e. after cutting), the growth of the survivors was correlated with variations in stand basal areas, while it was not the case at higher basal areas (i.e. prior to cutting). This suggests a significant enhancement of the survivor growth in these deciduous forests may only happen with a post-harvest basal area of 18 m² ha⁻¹ or less. Recommendations with respect to residual basal area and timber improvement rules are proposed which also take into consideration conclusions found in a literature review.

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Keywords: American beech; Dendrochronology; Diameter growth; Residual basal area; Stand growth and mortality; Sugar maple; Uneven-aged silviculture; Yellow birch

1. Introduction

The deciduous forest sub-zone in Southern Quebec (11 × 10⁶ ha) is composed largely of sugar maple (*Acer saccharum*, Marsh.) dominating uneven-aged stands. Since the mid-1980s, selection cutting has become the most important treatment applied in these stands reaching up to approximately 50,000 ha annually (Parent, 2002). Results from provincial experimental blocks over the last 15 years (Bédard and Majcen, 2001, 2003; Majcen and Richard, 1992) have shown that single-tree selection cutting can be sustainable under a regime of 30% removal with a 15–25 year cutting cycle. However, the monitoring by the Quebec Ministry of Natural Resources (MRN, now called MRNFP) 5 years after harvesting of commercially treated forests, revealed a much lower net growth than observed in the experimental blocks, mainly due to high mortality rates after harvest (Bédard and Brassard, 2002). The

authors explained the higher mortality mainly by the tree marking regime used by the forest companies, which was not compliant with the timber improvement requirements of the provincial regulations (Majcen et al., 1995). Bédard and Brassard (2002) recommended that cutting cycles should therefore be extended far beyond the 15–25 years set out in the Forest Management Manual (MRNFP, 2003). However, in their report, Bédard and Brassard (2002) pointed out a large variability in stand mortality and net growth across the province, suggesting that single-tree selection cutting can be sustainable in some stands but not in others.

More surveys are needed, especially on a regional basis, to estimate this variability in stand productivity and factors leading to contrasting productivities. Moreover, the lack of understanding of the factors affecting the stands' response to selection cutting (particularly after commercial rather than experimental cuttings) in Quebec has hampered the ability of silviculturists to make effective adjustments and has led to important uncertainties in regards to the allowable annual cut computations.

Growth and mortality following selection cutting is generally evaluated by re-measuring trees on a periodic basis

^{*} Corresponding author. Tel.: +1 819 983 6589; fax: +1 819 983 6588.

E-mail address: eforget@iqaff.qc.ca (E. Forget).

over a relatively long period of time (Nowak, 1996; Majcen et al., 2005). This method requires long lag periods and high precision diameter measurements prior to harvesting, a practice which is uncommon and very limited in commercial harvesting operations in Quebec. A relatively new and promising method, based on the chronological assessment of tree radial increments, allows the study of stands long after their harvesting without pre-harvest data. Indeed, dendrochronological techniques are used to study historical tree's growth rates (Jean and Bouchard, 1996; McKenzie et al., 2001; Jones and Thomas, 2004), fire history (Bergeron et al., 2001), insect outbreaks (Morin et al., 1993), air pollution (Innes and Cook, 1989), and to estimate year of death for trees (Storaunet and Rolstad, 2002; Brown et al., 1998; Daniels et al., 1997; Storaunet and Rolstad, 2004).

Using dendrochronological techniques on stands commercially harvested through single-tree selection cutting 10 years ago, this study aims to provide new information on stands' pre- and post-harvest growth and mortality response after commercial selection cutting. The specific objectives of this study are: (i) to reconstruct stand basal area pre- and post-harvesting, (ii) to evaluate species-specific growth response after cutting, and (iii) to assess stand productivity (i.e. gross growth and net growth) in response to harvesting.

2. Material and methods

2.1. Study area and experimental design

The study area (112 km²) included 12 uneven-aged stands (ranging from 9 to 45 ha) located about 100 km north of Montréal in southern Quebec (46°18'N, 74°14'W, Fig. 1). This region is characterized by broken topography made up of hills reaching 200 m in elevation with an average slope of 20% (Robitaille and Saucier, 1998). The area is part of the sugar maple-yellow birch bioclimatic domain (Gosselin and Saucier, 1999). Soils are characterized by thick till deposits (depth over 1 m), which was validated in all stands using a soil auger. Sites for sampling were randomly selected among stands dominated by sugar maple (SM) or sugar maple-yellow birch (SM-YB, *Betula alleghaniensis*, Britt.) and mapped as having the same density and height classes. All stands were over 80-years of age and had been harvested using single-tree selection cutting in 1992.

Within each of the 12 stands, three circular sample plots (625 m²) were randomly located while avoiding main skid trails in order to reduce sampling variability. However, in order to make a comparison, results were also calculated including main skid trails (cf. "basal area, growth and mortality assessments" in Section 2).

2.2. Data collection

In 2002, for all plots, living and dead stems with diameter-at-breast-height (DBH, i.e. 1.3 m above ground level) over 9.0 cm were measured using a calliper (mean of two perpendicular diameters). From 50% of randomly selected living trees, three

cores, each at 120° from each other, were taken at breast height. Depending on the species, 10 to 30 additional trees were cored in stands not harvested in recent years to be used as control stands. After analysis, the average annual growth for all three dominant species (i.e. sugar maple, yellow birch and American beech (*Fagus grandifolia*, Ehrh.)) was calculated for the 1988–2002 period. The diameter of each stump from harvested trees was measured 30 cm above ground level. For dead trees, a disc was taken at breast height, or higher on the bole when this provided a better quality disc. Two dominant living trees outside the plots, but within the stand, were felled and a disc taken at breast height in order to develop a master chronology for diameter growth.

2.3. Dendrochronological measurements

All samples were prepared following standard methods in dendrochronology (Swetnam et al., 1985). The ring-widths of the last 18 years for the cores of living trees, the last 50 years for the dead trees, and the last 80 years for the discs of the two living trees outside the plots were measured to the nearest 0.01 mm using a 40× magnification and a measuring device (Velmex Inc., Bloomfield, NY, USA) coupled with a digital meter. Each sample was measured dry, and to account for wood shrinking in the radial axis, a coefficient of 4.8% [calculated for hardwood species from Haygreen et al. (1982)] was applied to the measured growth.

Cross-dating was performed to validate measured series and to date tree death. Series of data from the same species were cross-dated using the COFECHA computer program (Holmes, 1997) and corrected for missing rings and measurement problems according to standard procedures (Swetnam et al., 1985). To make corrections, the program evaluates correlations between 30-year segments of each standardized series and a temporary mean standardized series. Each segment was moved on a 21-year window using 1-year steps. Correlation output gives the best position of each segment, pointing out the missing rings or extra measured rings, and results were visually validated. Using living trees, a master chronology was developed for sugar maple, yellow birch and American beech for the region of the study area and used to date tree death. Since no coniferous trees were available in the stands, a chronology of 18 white spruce from a near-by site (Mont-Tremblant, Que.) (Jardon, 2001) was used. Among the 90 discs extracted from dead trees, 48 discs were in good enough condition to allow ring measurement and dating. The chronology correlations of 88% of the 48 discs dated were significant at 99%.

2.4. Basal area, growth and mortality assessments

Basal area in 2002 (BA₂₀₀₂) was calculated from diameter measurements taken in 2002 (cf. Table 1). The diameter immediately after cutting (i.e. 1993) was estimated from dendrochronological measurements for each living and dead stem with a DBH over 9.0 cm in 2002. All living stems in 1993 with a DBH over 9.0 cm were used to evaluate the stands' basal area in 1993 (BA₁₉₉₃). Living stems in 2002 showing a DBH of

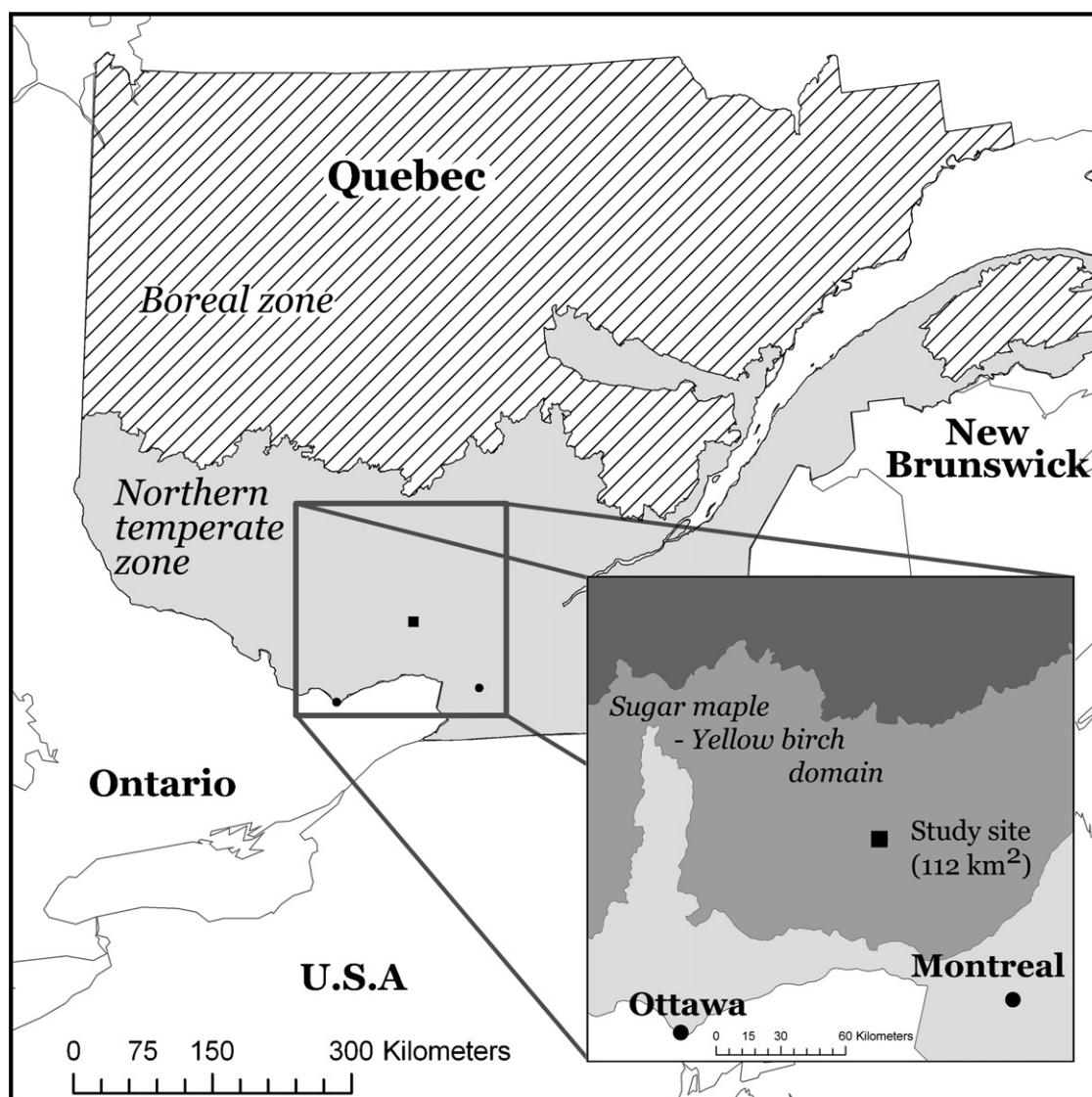


Fig. 1. Location of the study area in southern Quebec, Canada. The 12 studied stands were sampled within an area covering 112 km².

Table 1
Summary of the acronym, unit, definition and computation of the parameters used to characterize basal area and productivity of the 12 studied stands in southern Quebec

Name	Definition	Computed from
BA_{2002} (m ² ha ⁻¹) ^a	Basal area in 2002	Measured diameter in 2002
BA_{1993} (m ² ha ⁻¹) ^{a,b}	Basal area of living stems with a DBH over 9.0 cm in 1993 (after cutting)	Dendrochronological measurements
$BA_{removal}$ (m ² ha ⁻¹) ^{a,b}	Harvested basal area	Measured stump diameter in 2002 corrected for breast height
BA_{1992} (m ² ha ⁻¹) ^{a,b}	Basal area of living stems with a DBH over 9.0 cm in 1992 (prior to cutting)	$BA_{1993} + BA_{removal}$
$BAGR_{survivors}$ (m ² ha ⁻¹ year ⁻¹) ^a	Basal area growth rate of living trees in 1993 and 2002 with a DBH over 9.0 cm in both years divided by 10	Dendrochronological measurements
$BAGR_{ingrowth}$ (m ² ha ⁻¹ year ⁻¹) ^a	Basal area growth rate of living trees with a DBH over 9.0 cm in 2002 and under 9.1 cm in 1993 divided by 10	Dendrochronological measurements
$BAGR_{gross\ growth}$ (m ² ha ⁻¹ year ⁻¹) ^a	Stand basal area gross growth rate between 1993 and 2002 divided by 10	$BA_{survivor} + BA_{ingrowth}$
$BAGR_{mortality}$ (m ² ha ⁻¹ year ⁻¹) ^{a,b}	Basal area growth rate of living trees in 1993 with a DBH over 9.0 cm but dead in 2002 divided by 10	Dendrochronological measurements of dead trees
$BAGR_{net}$ (m ² ha ⁻¹ year ⁻¹) ^{a,b}	Stand basal area net growth rate between 1993 and 2002 divided by 10	$BA_{gross\ growth} - BA_{mortality}$

^a Parameter corrected to include main skid trails (for more details refer to Section 2).

^b Parameter susceptible to change depending on minimal and maximal mortality hypotheses (for more details refer to Section 2).

9.0 cm or lower in 1993 were labelled as ingrowth and not incorporated in the 1993's basal area calculations. The harvested basal area (BA_{removal}) of each stand was estimated using the stumps' basal area [corrected to breast height using stump height/breast height correction factors (MRNFP, 2005)]. Basal area prior to cutting (BA_{1992}) was assessed by adding BA_{removal} to BA_{1993} (Table 1).

To allow the comparison with studies that include main skid trails, the relative area covered with main skid trails was calculated and used to adjust estimated basal area, growth and mortality. To do this, the linear distance of skid trails was estimated using GIS software (ESRI ArcView 3.2), for the whole study area and by multiplying it by the average trail mean which was set at 5 m (E. Forget unpublished data). The area of main skid trails thus accounted for 4.8% of total treated forest area. When including skid trails, BA_{1993} and BA_{2002} were then estimated by applying a 0.952 coefficient. Beside, when including skid trails, estimations of BA_{removal} were computed from BA_{1992} for which 4.8% of the area was harvested totally (i.e. main skid trails) and 95.2% was harvested at the percentage found from computations made while excluding skid trails. For these calculations (as well as for basal area growth rate computations), the following three assumptions were made: (i) main skid trails were homogeneously distributed within the treated area and therefore the same correction coefficient could be applied to the 12 studied stands, (ii) 100% of the basal area was harvested in main skid trails, and (iii) no growth which could be incorporated in an inventory occurred in main skid trails during the 10 years following their establishment.

Mean annual basal area growth rates of survivors ($BAGR_{\text{survivor}}$) and ingrowth ($BAGR_{\text{ingrowth}}$) were calculated as the cumulated gain in basal area over the 10-year period of survivors (living trees in 1993 and 2002 with DBH over 9.0 cm in both years) and ingrowth trees (living trees with DBH over 9.0 cm in 2002 and under 9.1 cm in 1993) divided by 10 (Table 1). To include skid trails in the calculation, estimation of both parameters was achieved by applying the 0.952 skid trail reduction coefficient. Mean annual basal area gross growth (including skid trails or not) was calculated by adding the mean annual basal area's growth of survivors to that of the ingrowth (Majcen et al., 2005) (Table 1).

Mean annual diameter growth of survivors was also evaluated for the three dominant species for the total 15-year period and for each of the three consecutive 5-year periods. The first period corresponds to the 5 years of growth prior to the harvest (1988–1992) while the other two periods are post-harvest (1993–1997 and 1998–2002). For each species, calculation of the mean absolute diameter growth was achieved by subtracting the growth of the control plot trees from the growth of harvested plot trees. Relative diameter growth was computed by dividing the growth of sampled trees in harvested stands by the growth of sampled trees in the control plots.

For each stand, mean annual basal area mortality ($BAGR_{\text{mortality}}$) was calculated by summing the basal area of trees (alive in 1993 and with DBH over 9.0 cm) which died during the 10-year period, divided by 10 (Table 1). This $BAGR_{\text{mortality}}$ corresponded to the minimum measurable

mortality (thereafter called minimum mortality hypothesis). Indeed, considering that approximately half of the dead stem samples were not in good enough condition to allow ring measurement and dating, it is possible that mortality is underestimated. Alternatively, a maximum mortality rate was calculated by including the basal area of all dead samples to the calculation (thereafter called maximum mortality hypothesis), here assuming that all dead stems found in 2002 died in last 10 years. The skid trails reduction coefficient was also applied to both $BAGR_{\text{mortality}}$ parameters. Finally, net growth ($BAGR_{\text{net growth}}$) of each stand was calculated by subtracting the range of estimated mortality to the gross growth (Majcen et al., 2005) (Table 1). This allowed the calculation of a minimum and maximum net growth with the inclusion of main skid trails.

2.5. Data analyses

A repeated-measures analysis of variance (ANOVA) was used to compare growth among species (sugar maple, American beech and yellow birch) and for each of the three 5-year periods. Analyses of variance were used to determine significant differences in species relative proportions between forest types, while paired *T*-test were used to assess differences in 1992 and in 1993 between minimum and maximum basal area. A *T*-test was also used to determine differences between minimum and maximum $BAGR_{\text{mortality}}$ and between minimum and maximum $BAGR_{\text{net growth}}$. Linear regressions were established for the relationship between $BAGR_{\text{survivor}}$ before and after harvest. Finally, linear regressions were also performed between residual basal area (basal area after harvest) and $BAGR_{\text{gross growth}}$, $BAGR_{\text{mortality}}$, and $BAGR_{\text{net growth}}$ using data from a literature review of studies carried out in North America. All statistical analyses were performed using NCSS software (Hintze, 1999) and in all tests, significance level was set at $\alpha = 5\%$.

3. Results

3.1. Species composition and stand structure after selection cutting

Following harvest, sugar maple was the dominant species of each of the 12 studied stands accounting, on average, for 71% of the total basal area (Table 2). Other dominant species were American beech and yellow birch, representing, respectively, 15 and 9% (Table 2). Other species such as balsam fir (*Abies balsamea*, (L.) Mill.), red spruce (*Picea rubens*, Sarg.), black ash (*Fraxinus nigra*, Marsh.), red maple (*Acer rubrum*, L.), quaking aspen (*Populus tremuloides*, Michx.) and eastern white cedar (*Thuja occidentalis*, L.) represented 2% or less of stand species composition (data not shown). No significant differences were found in the relative proportion of the three main species between the two stand classifications (i.e. SM and SM-YB) (Table 2).

Stem frequency among the five DBH classes were similar for all 12 studied stands (Table 2) and was not significantly different between SM and SM-YB stands (Table 2), except for

Table 2

Proportion of the three dominant species as a function of stand total basal area immediately after harvest and stem frequencies among five DBH classes

Forest types	Stand	SM (% of total BA)	YB (% of total BA)	AB (% of total BA)	DBH classes (cm)				
					10–19	20–29	30–39	40–49	>50
SM-YB	1	48	21	9	0.37	0.37	0.20	0.03	0.03
SM	2	52	9	39	0.50	0.16	0.16	0.09	0.09
SM	3	87	1	9	0.45	0.21	0.18	0.09	0.06
SM-YB	4	85	9	0	0.47	0.41	0.12	0.00	0.00
SM-YB	5	49	32	5	0.32	0.48	0.13	0.03	0.03
SM-YB	6	93	5	2	0.39	0.19	0.14	0.17	0.11
SM-YB	7	61	5	35	0.42	0.33	0.12	0.06	0.06
SM-YB	8	82	9	10	0.44	0.39	0.11	0.00	0.06
SM	9	75	2	22	0.38	0.32	0.21	0.03	0.06
SM	10	78	5	4	0.38	0.41	0.16	0.00	0.06
SM	11	49	14	36	0.48	0.21	0.24	0.03	0.03
SM	12	91	0	9	0.42	0.22	0.14	0.14	0.08
SM mean	<i>n</i> = 6	72 a	5 a	20 a	0.44 a	0.26 a	0.18 a	0.06 a	0.06 a
SM-YB mean	<i>n</i> = 6	70 a	14 a	10 a	0.40 a	0.36 a	0.14 b	0.05 a	0.05 a
Average	<i>n</i> = 12	71	9	15	0.42	0.31	0.16	0.06	0.06

Results are presented for the 12 sugar maple dominated stands representing two forest types, sugar maple (SM) and sugar maple-yellow birch (SM-YB). Contrasting letters (a and b) correspond to significant differences between forest types at $\alpha = 5\%$. SM: sugar maple; YB: yellow birch; AB: American beech.

the 30–39 cm class (*T*-test, $p = 0.0485$). When averaged for all stands, stand structure thus corresponded to the classic inverse *J* curve (Table 2).

3.2. Changes in stand basal area

Estimates of the BA_{1992} and BA_{1993} of the 12 stands were very similar whether or not main skid trails were included (data not shown). As a consequence, results will mainly focus on calculations including main skid trails. *T*-test results showed BA_{1992} and BA_{1993} differed when the minimal measurable or maximal possible mortality hypotheses were used. BA_{1992} significantly varied between stands from 19.6 to 29.0 $m^2 ha^{-1}$, averaging 22.5 $m^2 ha^{-1}$ when the minimal measurable mortality

assumption was used and from 21.6 to 30.5 $m^2 ha^{-1}$, averaging 24.1 $m^2 ha^{-1}$ using the maximal possible mortality assumption (Table 3).

The proportion of basal area that was commercially harvested varied from 21.7 to 42.1% among studied stands (Table 3). The average harvested proportion of basal area ($BA_{removal}$) reached 29.6% of pre-cut basal area (i.e. 6.9 $m^2 ha^{-1}$) when skid trails were included (Table 3), being slightly higher than, but not significantly different from, the 26% (i.e. 6.1 $m^2 ha^{-1}$) estimated with exclusion of main skid trails (data not shown). At the same time, however, while skid trails corresponded to only 4.8% of total stand area, they accounted for 15.9% of the removal. Since a similar proportion of BA was harvested in each stand, estimates of BA_{1993}

Table 3

Maximal and minimal pre- (BA_{1992}) and post- (BA_{1993}) cut basal area ($m^2 ha^{-1}$) and removal ($BA_{removal}$) with main skid trails included

Stand	BA_{1992}		$BA_{removal}$				BA_{1993}	
	Max ($m^2 ha^{-1}$)	Min ($m^2 ha^{-1}$)	In stand ($m^2 ha^{-1}$) ^a	In skid trails ($m^2 ha^{-1}$) ^a	Total ($m^2 ha^{-1}$) ^a	Total (% of BA_{1992}) ^a	Max ($m^2 ha^{-1}$)	Min ($m^2 ha^{-1}$)
1	22.5	21.0	4.4	1.0	5.4	24.9	17.1	15.6
2	23.5	22.0	6.7	1.1	7.8	34.1	15.7	14.2
3	22.4	21.4	5.8	1.1	6.9	31.3	15.5	14.5
4	21.6	20.4	5.0	1.0	6.0	28.4	15.6	14.4
5	21.9	19.6	4.6	1.0	5.6	26.8	16.3	14
6	25.0	25.0	5.9	1.2	7.1	28.4	17.9	17.9
7	22.1	19.8	5.0	1.0	6.0	28.4	16.1	13.8
8	22.6	22.3	8.4	1.1	9.5	42.1	13.1	12.8
9	25.9	24.4	5.6	1.2	6.8	27.1	19.1	17.6
10	30.5	29.0	8.8	1.4	10.2	34.2	20.3	18.8
11	23.9	22.5	5.2	1.1	6.3	27.4	17.6	16.2
12	27.9	22.9	4.3	1.2	5.5	21.7	22.4	17.4
Average	24.1 a	22.5 b	5.8	1.1	6.9	29.6	17.2 a	15.6 b

Maximal and minimal basal area values were obtained when minimal or maximal mortality hypotheses were used for computations. Contrasting letters (a and b) refer to significant differences in the paired *T*-test when maximal and minimal hypotheses of mortality are used for computations.

^a Values presented for $BA_{removal}$ correspond to the mean of the values obtained with the minimal and maximal mortality hypotheses.

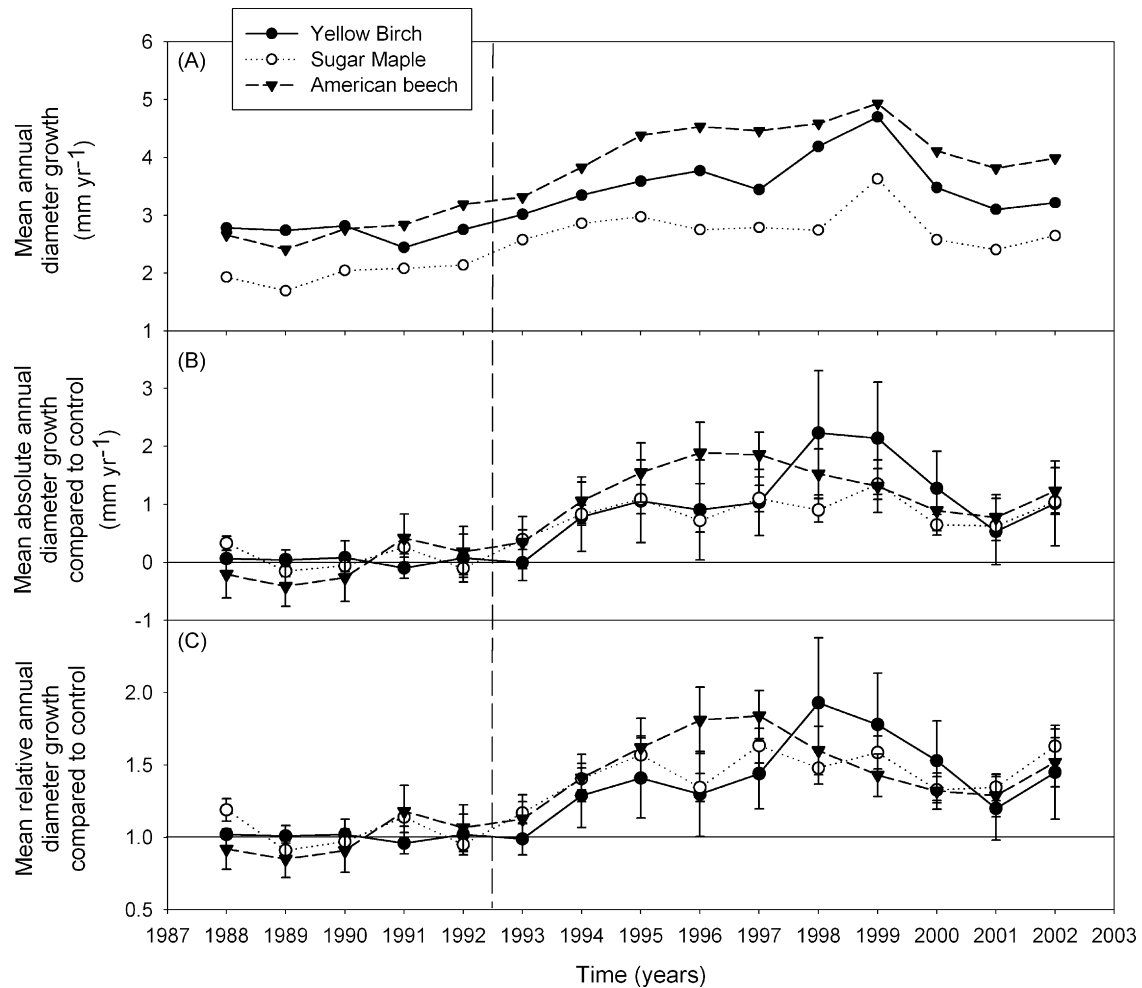


Fig. 2. Mean annual diameter growth (A) and mean absolute (B) and relative (C) diameter growth compared to control for yellow birch (filled circles–straight line), sugar maple (open circles–dotted line) and American beech (filled down triangles–dashed line). The vertical dashed line indicates when the selection cutting was performed.

followed patterns very similar to those of BA₁₉₉₂. On average, BA₁₉₉₃ reached 15.6 and 17.2 m² ha⁻¹ when including main skid trails (Table 3).

3.3. Species growth response to selection cutting

Among the three species, American beech showed the highest post-cut mean annual diameter growth for every year of the 10-year period, followed by yellow birch and sugar maple (Figs. 2A and 3A), with 10-year averages for each species being, respectively, 4.2, 3.6 and 2.8 mm year⁻¹. All three species responded rapidly (within 2 years) after selection cutting through an increase in mean annual diameter growth, but they showed contrasting patterns of response (Figs. 2B and C and 3B and C). American beech showed a rapid growth in the first 5-year period after harvest, while yellow birch's growth was highest for the last 5-year period (Figs. 2 and 3, Table 4). Sugar maple diameter growth remained relatively constant during the 10 years following harvest (Figs. 2 and 3, Table 4). However, when comparing absolute and relative growth averages for the 5-year periods after harvest, no differences

between species and between post-harvest periods were reported (Fig. 3, Table 4). As a result, post-harvest (1993–2002) diameter growth increased over control growth by 0.8, 1.0 and 1.2 mm year⁻¹, respectively, for yellow birch, sugar maple and American beech. This corresponded to relative increases, respectively, of 44, 45 and 50%; however, neither the absolute nor the relative growth values were statistically different from each other (Fig. 3, Table 4).

3.4. Stand growth and mortality

Similar to basal area findings, the effect of including or excluding main skid trails made little difference in estimates of all BAGR parameters. Thus, for clarity and to allow comparison with other studies (cf. Table 5), further result descriptions focus on those obtained with the inclusion of skid trails.

BAGR_{survivors} reached 0.36 m² ha⁻¹ year⁻¹ (S.E. = 0.01), while BAGR_{ingrowth} reached 0.05 m² ha⁻¹ year⁻¹ (S.E. = 0.01) in the study area. Resulting BAGR_{gross growth} was consequently 0.41 m² ha⁻¹ year⁻¹ (S.E. = 0.02) (Fig. 4). Minimal and maximal limits for BAGR_{mortality} were 0.04 (S.E. = 0.01) and

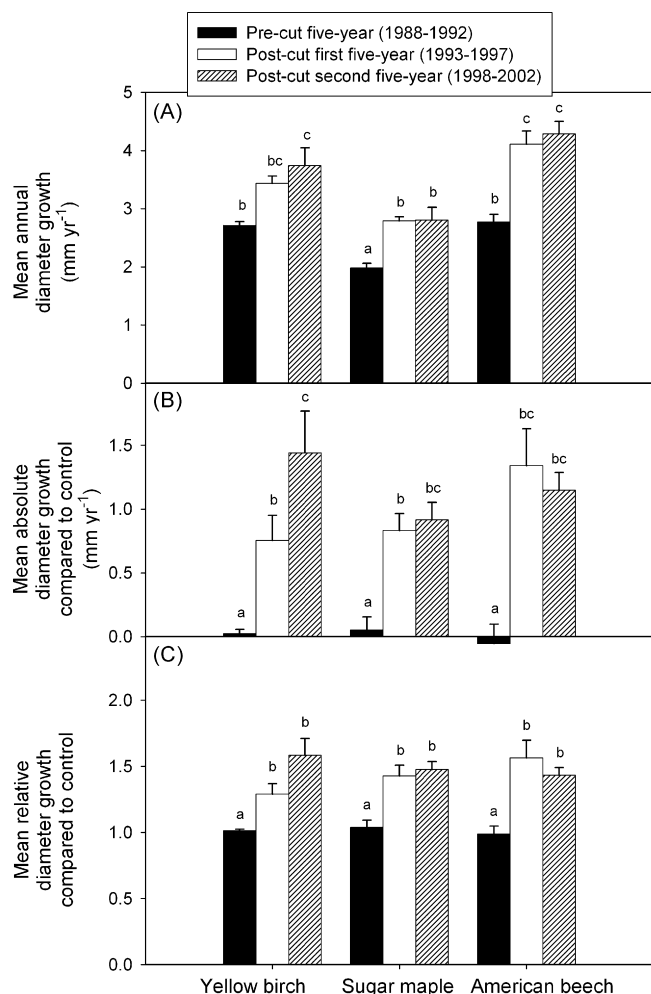


Fig. 3. Means and standard errors for annual diameter growth (A) and absolute (B) and relative (C) diameter growth compared to control for yellow birch, sugar maple and American beech. Results are averaged for three consecutive 5-year periods; the one prior to selection cutting (1988–1992; filled bars) and the first (1993–1997; open bars) and second (1998–2002; diagonal lines) ones following selection cutting. Contrasting letters refer to significant differences at $\alpha = 5\%$ in the repeated-measure ANOVA.

$0.20 \text{ m}^2 \text{ ha}^{-1} \text{ year}^{-1}$ (S.E. = 0.04), respectively, leading to a minimal and a maximal BAGR_{net} of 0.21 (S.E. = 0.02) and $0.37 \text{ m}^2 \text{ ha}^{-1} \text{ year}^{-1}$ (S.E. = 0.02), respectively (Fig. 5). For both $\text{BAGR}_{\text{mortality}}$ and BAGR_{net} , significant differences were reported between minimal and maximal limits in the T -test analysis (Fig. 5).

Table 4

F and p results for the repeated-measures ANOVA for differences between mean annual diameter growth and mean annual absolute and relative diameter growth (as compared to control tree growth) as a function of species, three contrasting five-year periods and their interaction

Sources of variation	d.f.	Mean annual diameter growth		Mean absolute annual diameter growth compared to control		Mean relative annual diameter growth compared to control	
		F	p	F	p	F	p
Species	2	35.09	<0.0001	0.97	0.3900	0.11	0.8918
Five-year periods	2	34.96	<0.0001	32.41	<0.0001	30.36	<0.0001
Interaction	4	1.35	0.2694	1.95	0.1234	1.82	0.1466

3.5. Stand basal area and mean annual growth

The relationship between pre- and post-cut basal area and pre- and post-cut mean annual basal area growth for survivors showed that the decrease in basal area due to selection cutting increased survivors' growth for all stands (Fig. 6). However, the relationship between stands' basal area and growth differed between the pre- and post-cut stand population. Variations in the mean annual basal area growth of survivors (open circles in Fig. 6) were independent of basal area in pre-cut stands (i.e. in a range of higher stand basal areas) while this was not the case in the post-cut population (i.e. lower stand basal area) (filled circles in Fig. 6).

4. Discussion

4.1. Stand historical reconstruction

On an area basis ($\text{m}^2 \text{ ha}^{-1}$), including skid trails slightly increased the basal area harvested and thus decreased the estimated post-cut stand basal area. Even though main skid trails account for less than 5% of the harvested area, including them in the calculations caused the removal to increase by 14%. Because of the high between-stand variability in pre-cut stand basal area and in the percentage of the basal area harvested, differences caused by main skid trails can thus be considered as minor. Indeed, no significant differences were found for any parameters including skid trails or not (data not shown). However, care has to be taken when extrapolating these results in larger time or space scales, where these differences might become critical. Furthermore, one should acknowledge that the proportion of skid trail area at the landscape or management unit level might differ from the 4.8% reported in this study. Depending on the prescription, the proportion of skid trail area may be larger and potentially result in significant differences in BA estimates even at the stand scale.

4.2. Survivor's growth response to selection cutting

The effect of selection cutting (i.e. a removal of 29.6% of pre-cut basal area) significantly increased survivor's growth for the three main species, with a response occurring within the first 2 years after canopy opening (Fig. 2). Jones and Thomas (2004) found similar results for sugar maple in Central Ontario. This rapid diameter (but not height) growth response after thinning was already associated with the need of individual trees to

Table 5

Summary of a literature review of the effect of selection cutting in eastern North American Hardwood forest

Source	Region	Post-cut basal area (m ² ha ⁻¹) ^a	BAGR _{gross growth} (m ² ha ⁻¹ year ⁻¹) ^a	BAGR _{mortality} (m ² ha ⁻¹ year ⁻¹) ^a	BAGR _{net} (m ² ha ⁻¹ year ⁻¹) ^a	Period (year)	Harvest type ^b
Crow et al. (1981)	MI (USA)	11	0.52	0.04	0.48	10	EXP
Majcen and Richard (1992)	QC (Canada)	12	0.43	0.02	0.41	4	EXP
Majcen (1995)	QC (Canada)	13	0.47	0.06	0.41	10	EXP
Erdmann and Oberg (1973)	WI (USA)	14	0.77	0.07	0.71	15	EXP
Majcen and Richard (1992)	QC (Canada)	14	0.42	0.25	0.18	4	EXP
Solomon (1977)	NH (USA)	14	0.58	0.03	0.53	10	EXP
Majcen (1995)	QC (Canada)	15	0.47	0.14	0.33	10	EXP
Miller (1997)	WV (USA)	16	0.79	0.05	0.74	10	EXP
Crow et al. (1981)	MI (USA)	16	0.52	0.05	0.47	10	EXP
Bédard and Majcen (2001)	QC (Canada)	17	0.44	0.12	0.32	10	EXP
Erdmann and Oberg (1973)	WI (USA)	17	0.71	0.08	0.62	15	EXP
Forget et al. (this study) ^c	QC (Canada)	17	0.41	0.04	0.37	10	COM
Majcen and Richard (1992)	QC (Canada)	18	0.47	0.10	0.37	4	EXP
Solomon (1977)	NH (USA)	18	0.49	0.11	0.38	10	EXP
Majcen (1995)	QC (Canada)	19	0.54	0.13	0.42	10	EXP
Bédard and Brassard (2002)	QC (Canada)	19	0.43	0.32	0.11	5	COM
Bédard and Brassard (2002)	QC (Canada)	20	0.41	0.29	0.12	5	COM
Bédard and Majcen (2001)	QC (Canada)	20	0.48	0.15	0.34	10	EXP
Miller (1997)	WV (USA)	21	0.81	0.06	0.75	10	EXP
Erdmann and Oberg (1973)	WI (USA)	21	0.71	0.10	0.61	15	EXP
Crow et al. (1981)	MI (USA)	21	0.46	0.06	0.40	10	EXP
Solomon (1977)	NH (USA)	22	0.50	0.22	0.28	10	EXP
Ondro and Love (1979)	ON (Canada)	22	0.39	0.12	0.27	10	EXP
Majcen (1995)	QC (Canada)	25	0.45	0.22	0.24	10	EXP
Majcen and Richard (1992)	QC (Canada)	25	0.46	0.23	0.24	4	EXP
Miller (1997)	WV (USA)	27	0.82	0.07	0.75	10	EXP

Presented data correspond to post-cut basal area (m² ha⁻¹), stand mortality (m² ha⁻¹ year⁻¹), stand gross growth (m² ha⁻¹ year⁻¹), stand net growth (m² ha⁻¹ year⁻¹), period of the survey (year) and type of harvest studied.

^a Means might have been performed for studies which reported results for stands with similar post-cut basal area.

^b Data collected from: COM: commercial logging operations; EXP: experimental study.

^c Data using the minimal mortality hypothesis (for details, refer to Section 2).

reconstruct an efficient water transport structure adapted to their new environmental conditions (Kneeshaw et al., 2002). Post-cut survivors' growth of all three species responded similarly by a 45–50% increase. However, since sugar maple showed the lowest pre-cut growth, its absolute diameter growth was lower (even though not significantly) than those of the two other species. The lower growth response of sugar maple compared to yellow birch and American beech might be related

to the widely observed decline of sugar maple in eastern North America. For example, Juice et al. (2006) showed that mature trees and seedlings of sugar maple in New Hampshire (USA) were significantly affected by a lack of calcium caused mainly

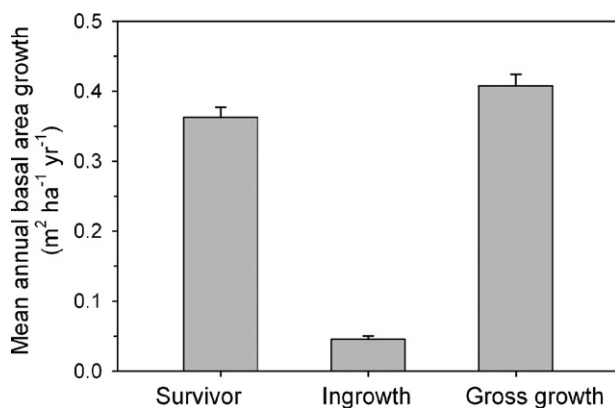


Fig. 4. Means and standards error of annual basal area growth (m² ha⁻¹ year⁻¹) for BAGR_{survivors}, BAGR_{ingrowth} and BAGR_{gross growth}. Results are presented with main skid trails included in computations.

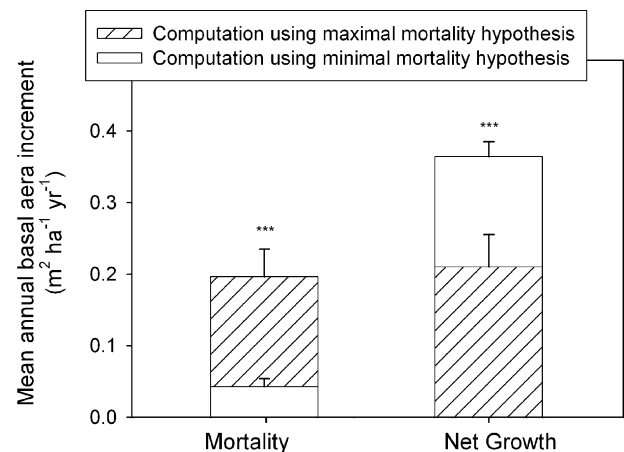


Fig. 5. Means and standards error of annual basal area mortality and net growth with main skid trails included in the computation. Results are presented for maximal (open bars) and minimal (diagonal lines) mortality hypotheses were used in computations. Indicated levels of significance (ns, *, **, ***) refer to the results of the *T*-test analysis comparing minimum and maximum mortality hypotheses.

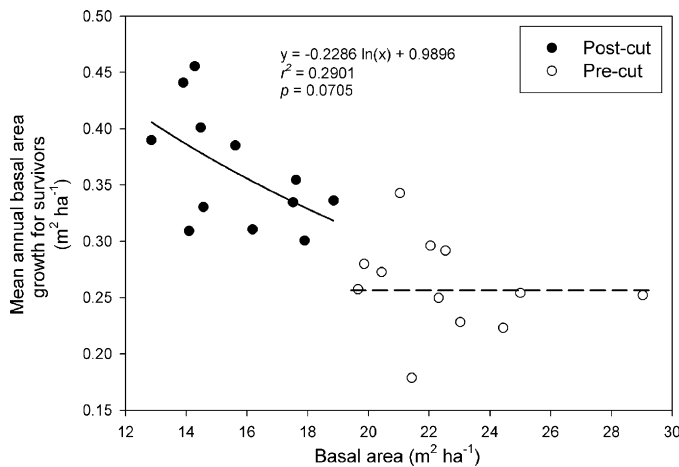


Fig. 6. Relationship between mean annual basal area growth for stand survivors ($\text{m}^2 \text{ha}^{-1}$) and stand basal area ($\text{m}^2 \text{ha}^{-1}$). Results are presented for pre-cut (open circles) and post-cut (filled circles) stand conditions. Best curve fitting for post-cut relationship was: $y = -0.2286 \ln(x) + 0.9896$, $r^2 = 0.2901$, while pre-cut relationship was not significant ($y = 0.26$).

by pollution and acid deposition. In addition, American beech has already been suspected to better use partial canopy disturbance regimes compared to sugar maple, especially close to the northern limit of sugar maple distribution area (Nolet et al., submitted). According to the authors, this was mainly related to American beech's better shade tolerance and a more efficient use of local poorer soils.

For all three species, results clearly showed that improved environmental conditions resulting from selection cutting lasted for all 10 sampled years following harvest (i.e. during the two 5-year periods following selection cutting), leading to significantly higher post-cut growth compared to pre-cut growth (Fig. 3, Table 4). Thus, while better increment growth has already been reported for the first 5 years after thinning (Simard et al., 2004; Jones and Thomas, 2004), our results suggest that on relatively fertile sites with openings representing about 30% of the canopy, growth enhancement is likely to be sustained for at least 10 years after harvest.

At the species level, the post-cut diameter growth obtained for yellow birch in this study (3.6 mm year^{-1}) was slightly higher than growth observed by Zarnovican and Laberge (1994) ($1.6\text{--}3.4 \text{ mm year}^{-1}$) in Southeastern Quebec after light thinning of dominant and codominant trees, and much higher than the value found by Roberge (1975) (2.1 mm year^{-1}). In contrast, the observed growth of sugar maple in this study (2.8 mm year^{-1}) was lower than results obtained by Crow et al. (1981) (4.1 mm year^{-1}) in a stand with a residual basal area of approximately 16 m^2 . This variability (within the same species) reported through the literature clearly reinforces the need to define species growth potential at a regional level, especially with the growing context of modelling stand productivity. When compared with the growth used in this region by the MRNFP for the computation of the allowable annual cut (MRNFP, 2003), slightly lower growth was found for sugar maple (2.8 mm year^{-1} versus 3.0 mm year^{-1}) but yellow birch

and American beech growth were, respectively, 0.7 and 1.8 mm year^{-1} higher than in the MRNFP model (3.0 and 2.4 mm year^{-1} , respectively). Such differences may critically affect the predictions dealing with the evolution of stand productivity and even composition. This is particularly true in this instance, where sugar maple BA accounted for the majority of the post-harvest basal area (Table 2) but showed the lower growth response to selection cutting.

4.3. Stand growth and mortality after commercial selection cutting

The mean annual basal area gross growth ($\text{BAGR}_{\text{gross growth}}$: $0.41 \text{ m}^2 \text{ha}^{-1}$) is made up of $0.36 \text{ m}^2 \text{ha}^{-1}$ from survivor growth ($\text{BAGR}_{\text{survivors}}$) and $0.05 \text{ m}^2 \text{ha}^{-1}$ from ingrowth ($\text{BAGR}_{\text{ingrowth}}$). This value is comparable (cf. Table 5) to those reported at similar residual basal area in other studies from experimental blocks in Quebec (Majcen and Richard, 1992; Bédard and Majcen, 2001, 2003), and for commercial selection cutting in Quebec (Bédard and Brassard, 2002). However, $\text{BAGR}_{\text{gross growth}}$ obtained in this study is slightly lower than those obtained in northern United States (Erdmann and Oberg, 1973; Solomon, 1977; Crow et al., 1981; Miller, 1997), possibly because of the better climatic conditions (longer growing season, higher mean annual temperatures and higher radiation input) in more southern latitudes.

In this study, mortality was assessed by calculating minimal (observed) and maximal (potential) stand mortality with the confidence that actual mortality is closer to the minimal limit. Indeed, considering that, (1) about 1/3 of the readable discs are from trees that died before harvest, (2) the majority of the most rotten and decayed readable discs died before harvest, and (3) the decay level of the disc is correlated with the number of years since death (Cline et al., 1980), we are confident that the majority of unreadable discs were from trees that died prior to harvesting. However, to provide a rigorous range within which stand mortality actually is, we kept both estimations which should be considered as the lower and higher limits.

The relative low stand basal area mortality obtained (especially when considering the minimal limit $0.04 \text{ m}^2 \text{ha}^{-1} \text{year}^{-1}$) was expected since mortality generally is lower in thinned stands compared to un-thinned stands (Yang et al., 1998; Simard et al., 2004). Stand mortality was within the range of all other studies carried out on experimental plots (Table 5). However, it was much lower than the mortality rate reported by Bédard and Brassard (2002) while studying a large number of commercially harvested stands across the province of Quebec in 1995 and 1996 (Table 5). According to the authors, the higher mortality rates reported in their study is mainly the result of an injudicious choice of the harvested stems. However, it cannot be excluded that the mortality rate was increased by factors such as low stand quality, poor soil quality or inadequate stand structure for selection cutting. Considering that sampling in this study was conducted on relatively homogeneous soil conditions within the same geographical area, the variation of the mortality among the 12 studied stands (range 0 to $0.53 \text{ m}^2 \text{ha}^{-1}$) was unusually large,

particularly in comparison with the range of mortalities reported by other (Table 5). The same conclusion can be drawn from Bédard and Brassard (2002) data, as the calculated confidence interval for the annual mortality varied from 0.17 to 0.41 m² ha⁻¹. Such observations thus reinforce the critical importance of studying and interpreting stand productivity at a regional and even micro-regional scale rather than landscape or provincial scale.

Finally, the net growth found in this study (0.20–0.37 m² ha⁻¹) is close to the results reported in other experiments (Solomon, 1977; Majcen and Richard, 1992; Bédard and Majcen, 2001) for similar post-cut basal area.

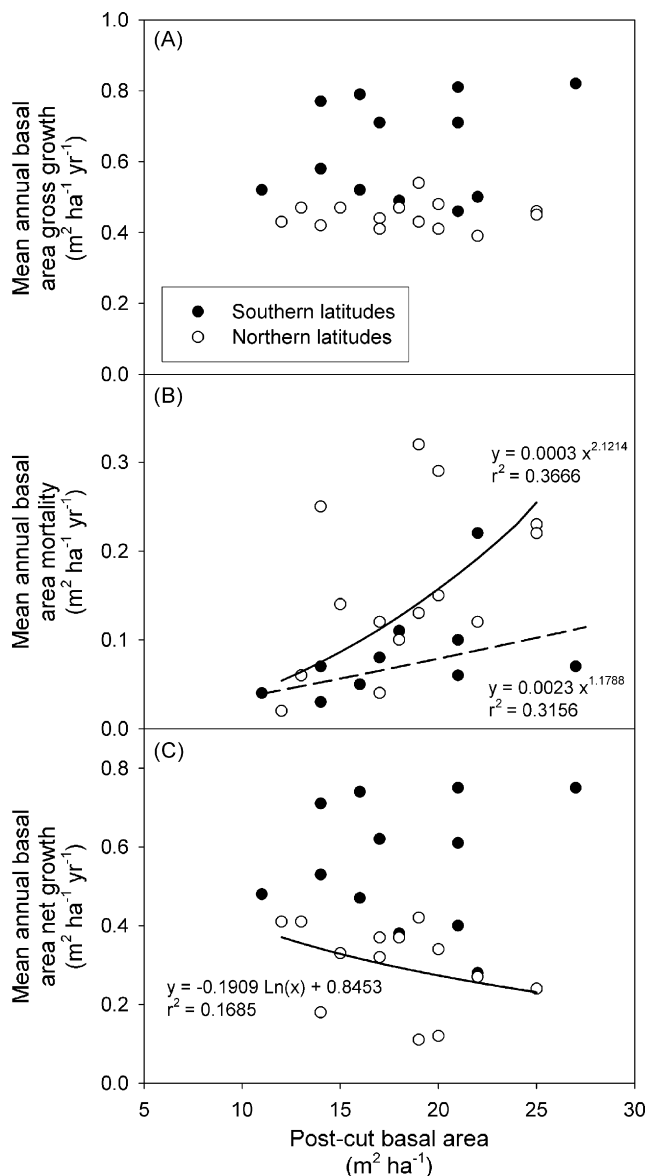


Fig. 7. Relationship between post-cut basal area and mean annual basal area gross growth (A), mortality (B) and net growth (C). Data come from the literature review reported in Table 5. Results are presented for studies carried out in the northern (open circles, i.e. Ontario and Quebec studies) and the southern (filled circles, i.e. northeastern US studies) latitudes of the north-eastern deciduous forest of North America. Only significant linear regressions are presented.

4.4. Stand productivity and basal area

As described in a 15-year survey by Erdmann and Oberg (1973), stand survivors' growth rates increased slightly at basal area between 9 and 16 m² ha⁻¹, were stable up to 18 m² ha⁻¹ and then decreased between 18 and 23 m² ha⁻¹. At the same time, between 9 and 23 m² ha⁻¹, ingrowth constantly increased while stand mortality constantly decreased. As a consequence, gross growth was relatively independent of the stand basal area as reported in Fig. 7. Indeed, at lower stand basal areas, the survivors' basal increment was high and ingrowth maximum. However, at higher basal areas, survivors' growth was lower and ingrowth was dramatically reduced, both because of higher levels of competition for space and resources. Thus, since mortality significantly increased with increasing stand BA, especially at northern latitudes (Fig. 7b), stand net growth was markedly reduced at high BA (Fig. 7c). From a stand productivity point of view, it will be more important at northern latitudes to maintain pre- and post-cut BA in a lower range (e.g. less than 22 m² ha⁻¹) to increase survivor growth and decrease stand mortality (Figs. 6 and 7). In contrast, maintaining stand BA in a higher range will critically lengthen cutting cycles. Similar conclusions dealing with a decrease in residual stand BA were as also reported by Niese and Strong (1992) and Niese et al. (1995).

It is noteworthy to mention that this study mainly focused on the relationship between stand productivity and stand residual basal area. However, additional factors such as crown size and shape, proportion of vigorous trees and species composition certainly play a key role in stand productivity variations, and more research is needed to quantify their respective importance.

5. Conclusion and implications for management

The large variation in mortality found among stands and its direct influence on net growth clearly highlights the importance of gaining a better understanding of the causes of mortality. This study showed that, in some specific cases, net growth can be increased by maintaining a relatively low basal area (e.g. 16–22 m² ha⁻¹) leading to shorter cutting cycles. This is realistic because keeping stands within these basal areas, especially in northern latitudes, will allow sufficient ingrowth, minimum mortality and maximum survivor growth (Figs. 6 and 7), which might not occur when keeping these stands within a higher range of basal areas (e.g. 20–28 m² ha⁻¹).

Our results also highlight the importance of multiplying similar studies to assess regional and micro-regional variability in stand productivity. Indeed, this is critical for a better understanding and prediction of larger scale land productivity. Using such studies to feed modeling tools, the response of different stand structures and compositions in relation to cutting intensity, post-cut BA and length of cutting cycles could be evaluated and consequently help to meet silvicultural and social objectives of forests.

Finally, one should also be concerned by the rapid and impressive growth response of American beech after selection cutting compared with sugar maple, which is observed to

respond less and even decline in certain areas. Further investigations on these issues are thus needed, especially because species composition and growth rates are essential elements of the annual allowable cut calculation.

Acknowledgements

Research funding for this project was provided by the MRNFP. The authors would like to thank M.-E. Drapeau and L. Tanguay, from Opti-Vert inc. for the field work. Thanks are also extended to S. Ostojic, J. Deslandes and D. Sénécal for the laboratory work, to co-workers D. Bouffard and F. Lorenzetti for their great contribution to this project and to S. Neamtz for the diligent editing. Constructive comments from C. Messier and two anonymous reviewers contributed to improve the manuscript to its final form.

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